

Descent of testes and ovaries

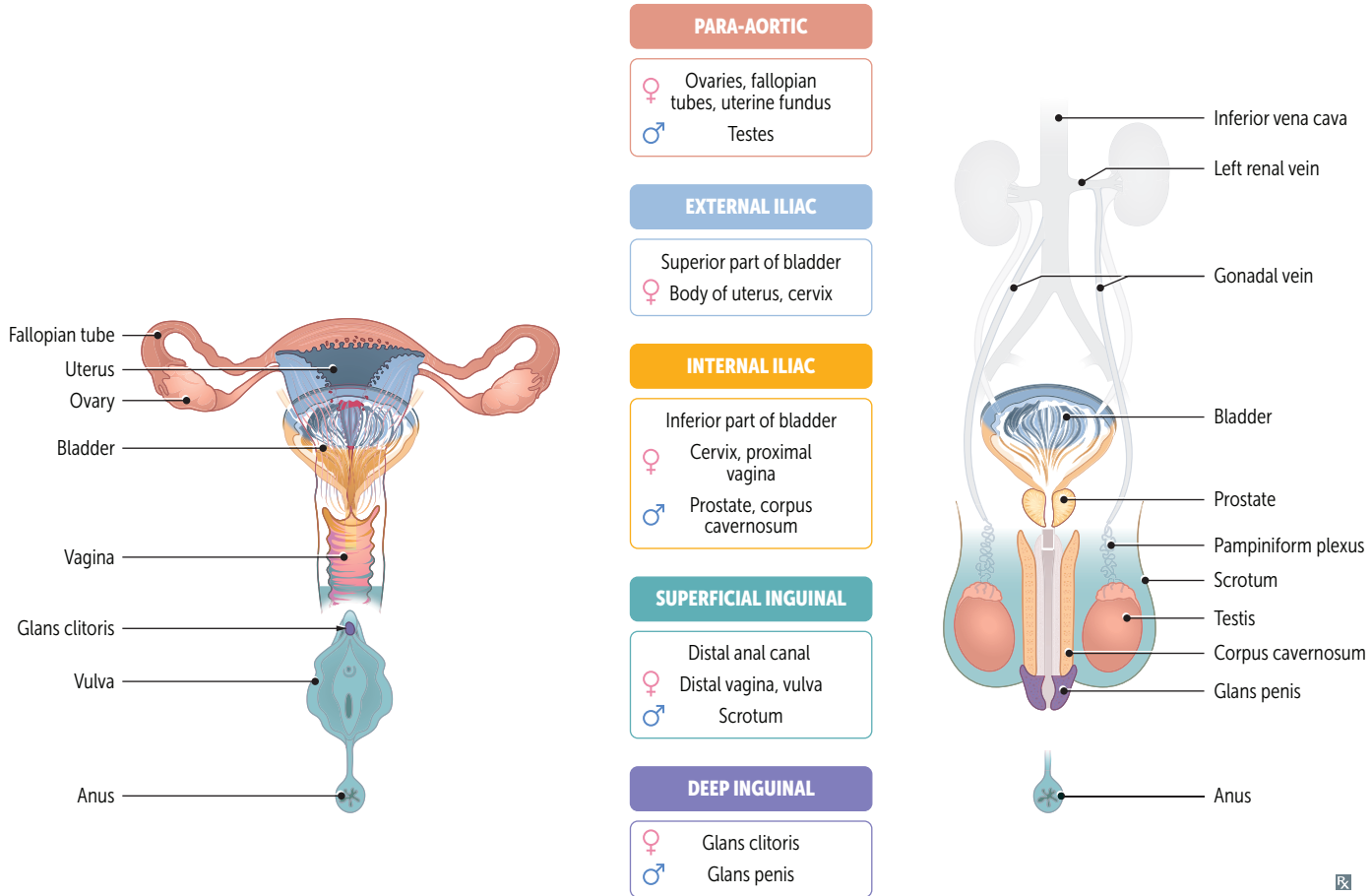
	DESCRIPTION	MALE REMNANT	FEMALE REMNANT
Gubernaculum	Band of fibrous tissue	Anchors testes within scrotum	Ovarian ligament + round ligament of uterus
Processus vaginalis	Evagination of peritoneum	Forms tunica vaginalis Persistent patent processus vaginalis → hydrocele	Obliterated

▶ REPRODUCTIVE—ANATOMY

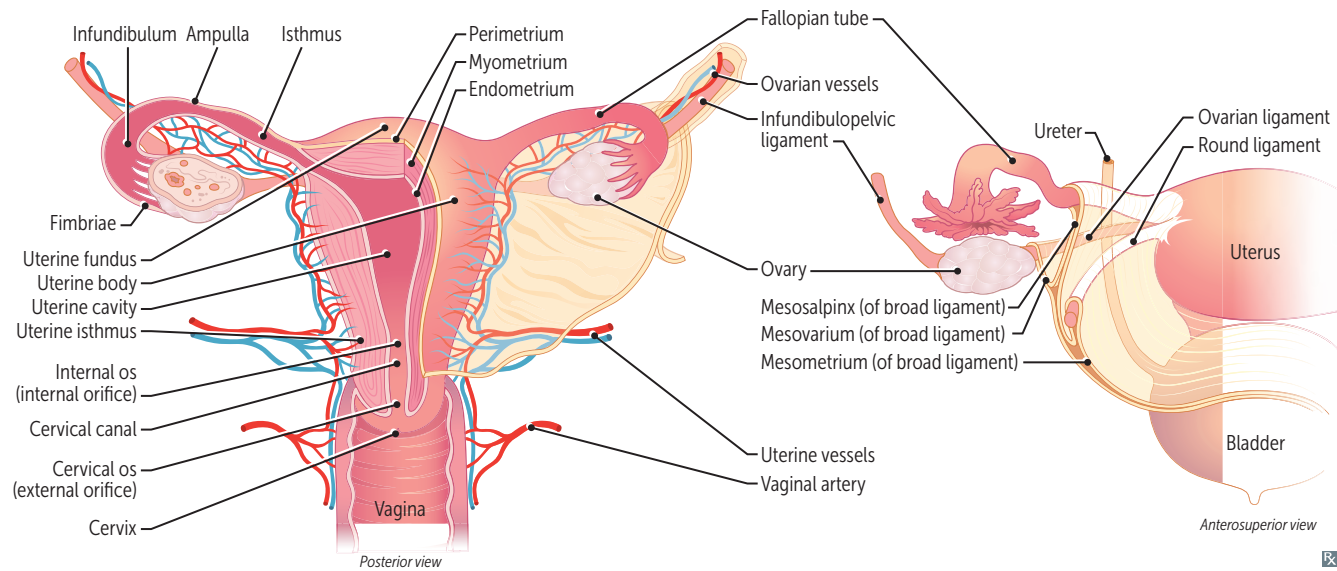
Drainage of reproductive organs

Venous drainage Right ovary/testis → right gonadal vein → IVC.
Left ovary/testis → left gonadal vein → left renal vein → IVC (takes the longer way).
Left testicular vein enters left renal vein at 90° angle → flow is less laminar on the left than on the right → left venous pressure > right venous pressure → varicocele is more common on the left.

Lymphatic drainage

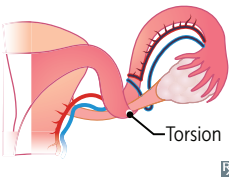


Female reproductive anatomy



LIGAMENT	CONNECTS	STRUCTURES CONTAINED	NOTES
Infundibulopelvic ligament	Ovary to lateral pelvic wall	Ovarian vessels	Also called suspensory ligament of ovary Ovarian vessel ligation during oophorectomy risks damaging the ureter
Utero-ovarian ligament	Ovary to uterine horn		Derivative of gubernaculum
Round ligament	Uterine horn to labia majora		Travels through inguinal canal Derivative of gubernaculum
Broad ligament	Uterus to lateral pelvic wall	Ovary, fallopian tube, round ligament	Fold of peritoneum comprising the mesometrium, mesovarium, and mesosalpinx
Cardinal ligament	Cervix to lateral pelvic wall	Uterine vessels	Condensation at the base of broad ligament Uterine vessel ligation during hysterectomy risks damaging the ureter
Uterosacral ligament	Cervix to sacrum		

Adnexal torsion



Twisting of ovary and fallopian tube around infundibulopelvic ligament and ovarian ligament → compression of ovarian vessels in infundibulopelvic ligament → blockage of lymphatic and venous outflow. Continued arterial perfusion → ovarian edema → complete blockage of arterial inflow → necrosis, local hemorrhage. Associated with ovarian masses/cysts. Presents with acute pelvic pain, adnexal mass, nausea/vomiting. Surgical emergency.

Pelvic organ prolapse

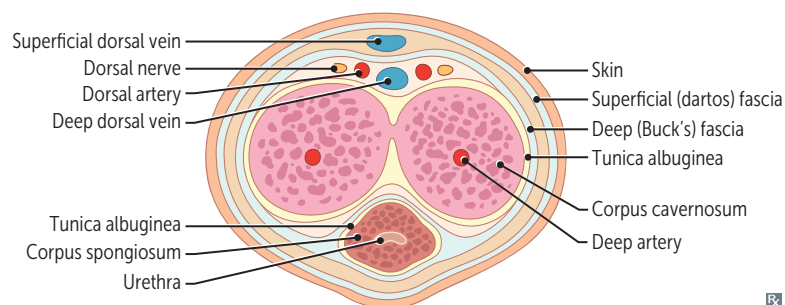
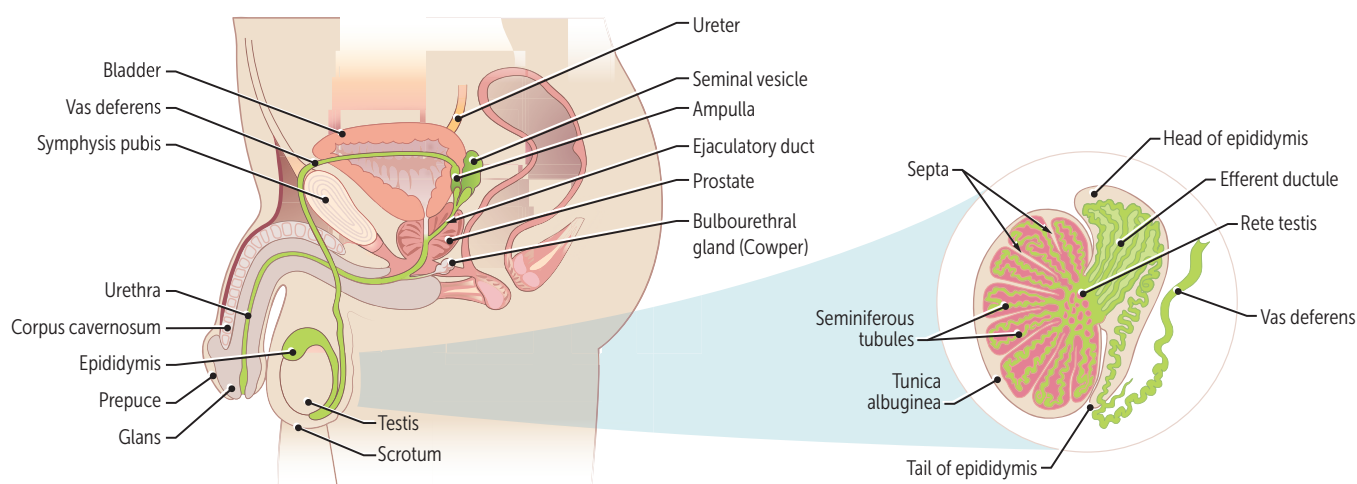
Herniation of pelvic organs to or beyond the vaginal walls (anterior, posterior) or apex. Associated with multiparity, ↑ age, obesity. Presents with pelvic pressure, bulging sensation or tissue protrusion from vagina, urinary frequency, constipation, sexual dysfunction.

- Anterior compartment prolapse—bladder (cystocele). Most common type.
- Posterior compartment prolapse—rectum (rectocele) or small bowel (enterocele).
- Apical compartment prolapse—uterus, cervix, or vaginal vault.

Uterine procidentia—herniation involving all 3 compartments.

Female reproductive epithelial histology

TISSUE	HISTOLOGY/NOTES
Vulva	Stratified squamous epithelium
Vagina	Stratified squamous epithelium, nonkeratinized
Ectocervix	Stratified squamous epithelium, nonkeratinized
Transformation zone	Squamocolumnar junction (most common area for cervical cancer; sampled in Pap test)
Endocervix	Simple columnar epithelium
Uterus	Simple columnar epithelium with long tubular glands in proliferative phase; coiled glands in secretory phase
Fallopian tube	Simple columnar epithelium, ciliated
Ovary, outer surface	Simple cuboidal epithelium (germinal epithelium covering surface of ovary)

Male reproductive anatomy

Pathway of sperm during ejaculation—

SEVEN UP:

Seminiferous tubules

Epididymis

Vas deferens

Ejaculatory ducts

(Nothing)

Urethra

Penis

Genitourinary trauma Most commonly due to blunt trauma (eg, motor vehicle collision).

Renal injury

Presents with bruises, flank pain, hematuria. Caused by direct blows or lower rib fractures.

Bladder injury

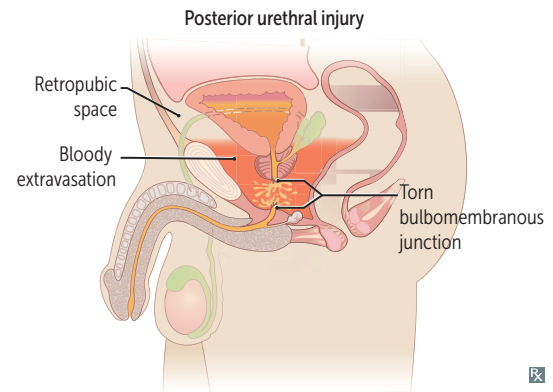
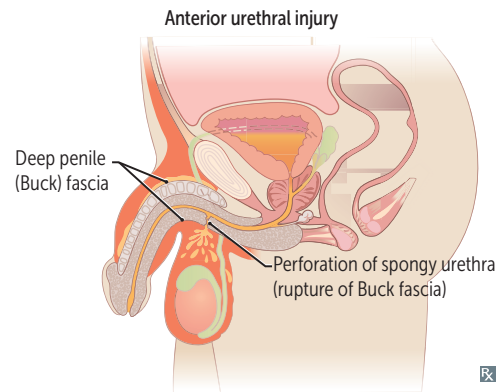
Presents with hematuria, suprapubic pain, difficulty voiding.

- Superior bladder wall (dome) injury—direct trauma to full bladder (eg, seatbelt) → abrupt ↑ intravesical pressure → dome rupture (weakest part) → intraperitoneal urine accumulation. Peritoneal absorption of urine → ↑ BUN, ↑ creatinine.
- Anterior bladder wall or neck injury—pelvic fracture → perforation by bony spicules → extraperitoneal urine accumulation (retropubic space).

Urethral injury

Occurs almost exclusively in males. Presents with blood at urethral meatus, hematuria, difficulty voiding. Urethral catheterization is relatively contraindicated.

- Anterior urethral injury—perineal straddle injury → disruption of bulbar (spongy) urethra → scrotal hematoma. If Buck fascia is torn, urine escapes into perineal space.
- Posterior urethral injury—pelvic fracture → disruption at bulbomembranous junction (weakest part) → urine leakage into retropubic space and high-riding prostate.



Autonomic innervation of male sexual response

Erection—**p**arasympathetic nervous system (pelvic splanchnic nerves, S2-S4):

- NO → ↑ cGMP → smooth muscle relaxation → vasodilation → proerectile.
- Norepinephrine → ↑ $[Ca^{2+}]_{in}$ → smooth muscle contraction → vasoconstriction → antierectile.

Emission—**s**ympathetic nervous system (hypogastric nerve, T11-L2).

Expulsion—visceral and **s**omatic nerves (pudendal nerve).

Point, **s**queeze, and **s**hoot.

S2, 3, 4 keep the penis off the **f**loor.

PDE-5 inhibitors (eg, sildenafil) → ↓ cGMP breakdown.

Seminiferous tubules

CELL	FUNCTION	LOCATION/NOTES
Spermatogonia	Maintain germ cell pool and produce 1° spermatocytes	Line seminiferous tubules A Germ cells
Sertoli cells	Secrete inhibin B → inhibit FSH Secrete androgen-binding protein → maintain local levels of testosterone Produce MIF Tight junctions between adjacent Sertoli cells form blood-testis barrier → isolate gametes from autoimmune attack Support and nourish developing spermatozoa Regulate spermatogenesis Temperature sensitive; ↓ sperm production and ↓ inhibin B with ↑ temperature	Line seminiferous tubules Non-germ cells Convert testosterone and androstenedione to estrogens via aromatase Sertoli cells are temperature sensitive, line seminiferous tubules, support sperm synthesis, and inhibit FSH Homolog of female granulosa cells ↑ temperature seen in varicocele, cryptorchidism
Leydig cells	Secrete testosterone in the presence of LH; testosterone production unaffected by temperature	Interstitium Endocrine cells Homolog of female theca interna cells

